

Probability & statistics

Properties of CDF

X be a r.v.

$$\forall t \in \mathbb{R} ; F(t) = \text{Prob}(X \leq t)$$

$$F: \mathbb{R} \rightarrow \mathbb{R}$$

i) $0 \leq F(t) \leq 1$

$$\forall t \in \mathbb{R}$$

ii) $\lim_{t \rightarrow \infty} F(t) = 1$

;

$$\lim_{t \rightarrow \infty^-} F(t) = 0$$

iii) F is a non-decreasing function.

$$\forall t_1, t_2 \in \mathbb{R}, t_1 \leq t_2 \Rightarrow F(t_1) \leq F(t_2)$$

iv) F is a right continuous function.

$$\text{for } \delta > 0, \quad \lim_{\delta \rightarrow 0} F(t + \delta) = F(t) \quad \forall t \in \mathbb{R}$$

 RHL = value of the function

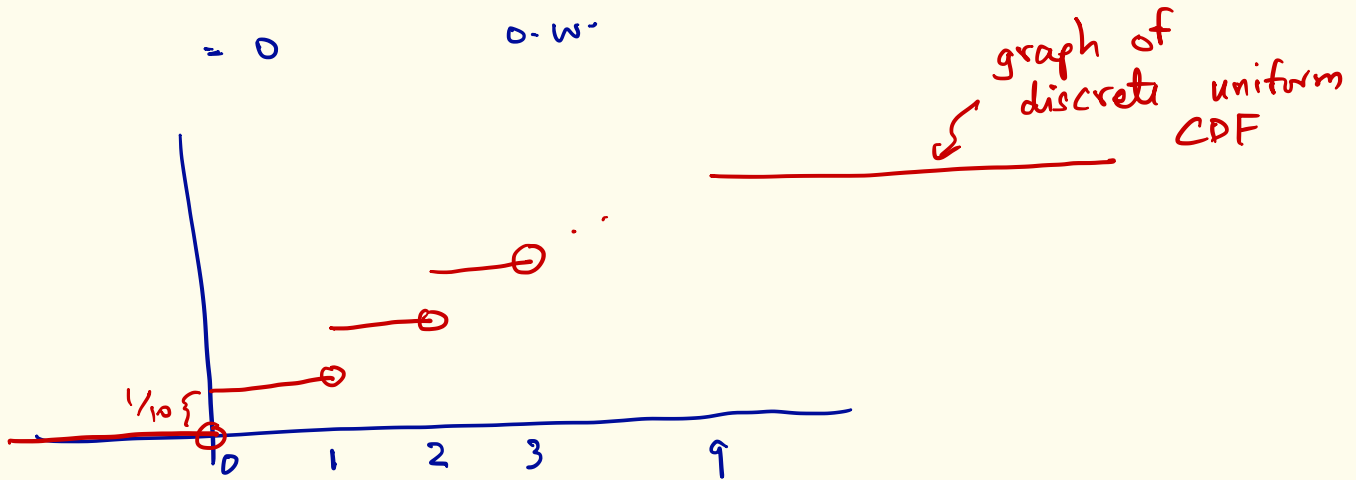
Observe: The function is discontinuous only at $t \in R_X$ when X is a discrete r.v.

Ex: $X \sim \text{uniform}(10)$

$$R_X = \{0, 1, 2, \dots, 9\}$$

$$f_X(x) = \frac{1}{10} \quad \text{for } x = 0, 1, 2, \dots, 9$$

$$= 0 \quad \text{o.w.}$$



A function $F: \mathbb{R} \rightarrow [0,1]$ is called a CDF (cumulative distribution function) if it satisfies above 4 properties.

From CDF to pmf (in case discrete r.v.)

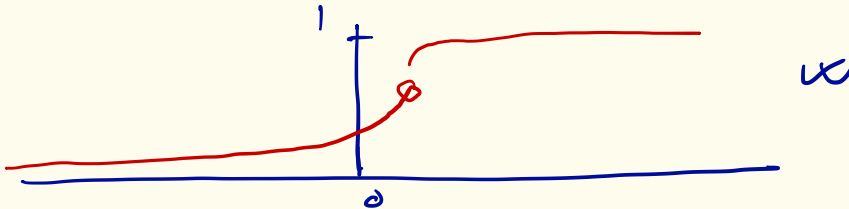
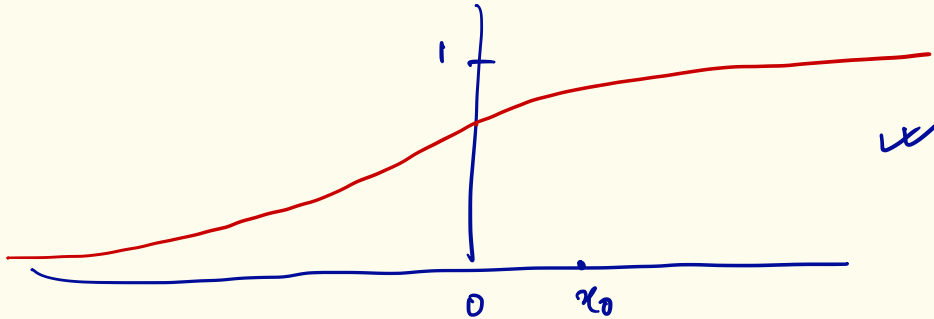
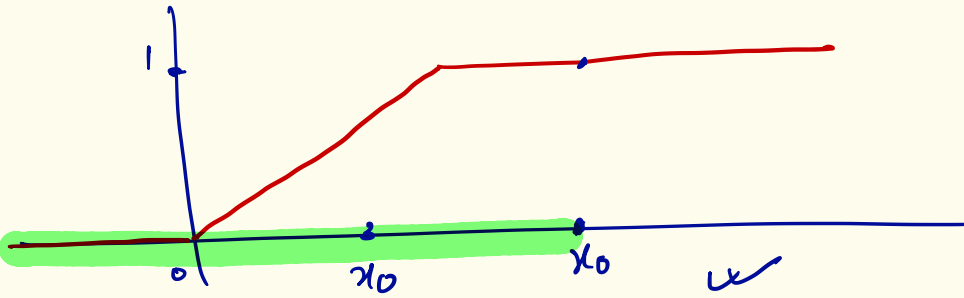
- check the pts in $\mathbb{R}$ where CDF has jumps.
- these are pts in the range of x .
- pmf at these pts is the jump size.

$\Omega, \mathcal{F}, P \rightarrow x \leftrightarrow \text{pmf} \leftrightarrow \text{CDF} \rightarrow$ We can go from
discrete
r.v.

CDF to pmf also
in case of discrete
r.v.

$$-\lim_{\delta \rightarrow 0^-} (F(x_0 + \delta) - \underline{F(x_0)})$$

$$= \boxed{\text{Prob}(X = x_0)}$$



Lebesgue decomposition theorem:

If a function $F(x)$ satisfies the properties (i) to (iv), (i.e. $F(x)$ is a CDF), then $F(x)$

can be represented as a sum of two F 's,

say $G(x)$ and $H(x)$ or

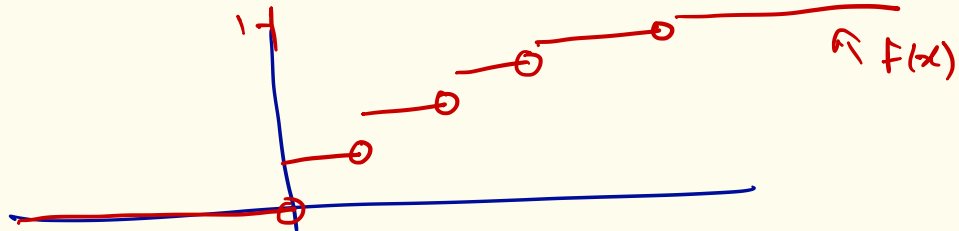
$$F(x) = G(x) + H(x) \rightarrow \text{continuous part} + \text{step function part}$$

where $G(x)$ is continuous and $H(x)$ is

a right continuous step function with jumps

coinciding with those of $F(x)$ at $H_x(-\infty) = 0$

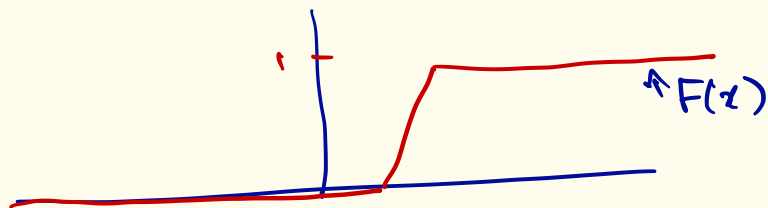
Ex:



$$G(x) = 0$$

$$H(x) = F(x)$$

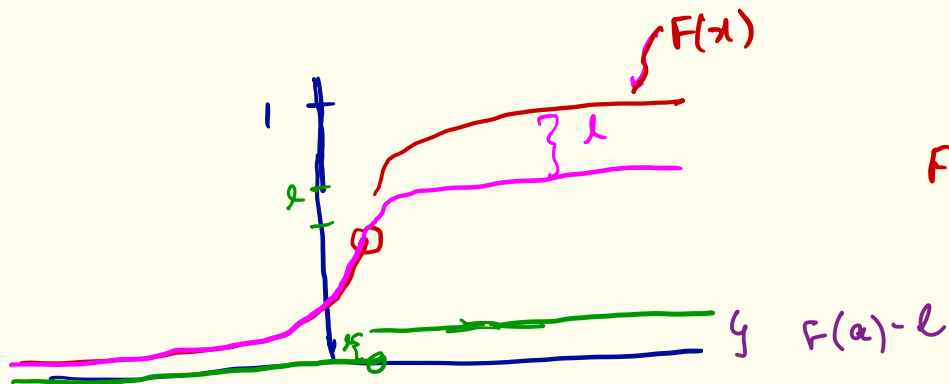
Ex:



$$G(x) = F(x)$$

$$H(x) = 0$$

Ex:



$$F(x) = G(x) + H(x)$$

$$y = F(x_0) - l$$

classification of r.v.s.

In Lebesgue decomposition thm,

- if $G(x) \equiv 0$, then the random variable is called as discrete r.v.
- if $H(x) \equiv 0$, then the random variable is called as continuous r.v.
- in the situation where neither $G(x)$ nor $H(x)$ is zero, we have a mixed r.v.

Aim: CDF $\rightarrow$ pmf $\rightarrow$ x
(or equivalent concept)

In case of discrete r.v.s.

CDF $\rightarrow$ pmf

by taking differences $F(t_i) - F(t_j)$

$t_i, t_j \in \mathbb{R}$

case of continuous r.v.

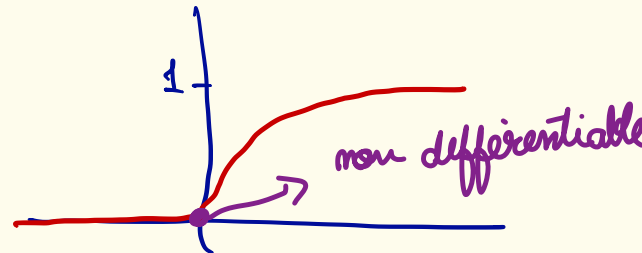
$$H(x) \equiv 0$$

$$P(X=x) = \lim_{\delta \rightarrow 0-} [F(x+\delta) - F(x)]$$

$\forall x \in \mathbb{R}$

$$= 0$$

Define $f(x) = \frac{d}{dx} F(x)$



note: The function F is differentiable almost everywhere

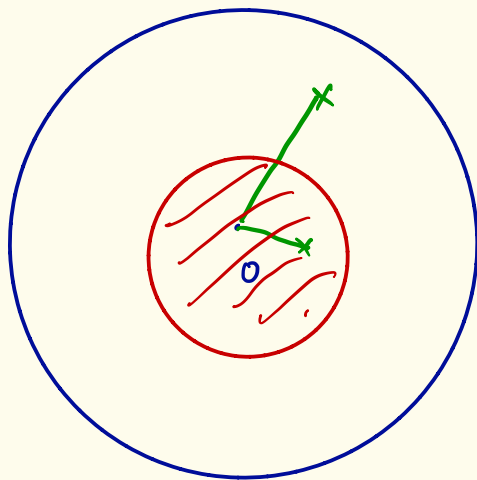
$\hookrightarrow$ points of non differentiability are "few"

gmp $f(x) = \frac{d}{dx} F(x)$

note: pdf does not give probability like pmf

↑
probability density function of a continuous r.v.

Ex:



Expt. Throw a dart on this circular board.

$(\Omega, \mathcal{F}, P)$

↑
uniform probability space.

Define
 x : distance of the dart from the centre of the board.

Let $F(x)$ denote the CDF of X .

$$F(x) = \text{Prob}(X \leq x)$$

$$= \frac{\pi x^2}{\pi R^2}$$

where

R : radius
of the board.

$$= \frac{x^2}{R^2}$$

$$F(x) = \begin{cases} 0 & x < 0 \\ x^2/R^2 & 0 \leq x \leq R \\ 1 & x > R \end{cases}$$

$$x < 0$$

$$0 \leq x \leq R$$

$$x > R$$